

Mark Raymond Takla

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TECHNICAL SKILLS AND LANGUAGES

Java | Python | C | R | React | ASP.NET | MATLAB | Haskell | Prolog | HTML | CSS | OrCAD Pspice | Linux | AI Tools

SQL | Pandas | NumPy | Scikit-Learn | PowerBI | TensorFlow | Keras | LangChain

Microsoft Word, PowerPoint, Excel | Adobe Photoshop, AfterEffects | UIPath

Arabic and English proficient | German A2 | French basics

EDUCATION

German University in Cairo (GUC)

Bachelor of Engineering

09/2023 - Current

Major in Computer Science

Cumulative German Scale GPA: 2.15 (equivalent to 3.3 (B+) American Scale GPA)

Cairo English School (CES) and New Ramses College (NRC)

Earlier Years

IGCSE

Early graduation after year 11 | Final score 110% with scale factor | 8 IGCSEs grade 9 (A* equivalent) | A Level mathematics grade A*

WORK EXPERIENCE

Bank NXT

Robotic Process Automation (RPA) Internship

07/2025 – 08/2025

- Got introduced to the banking industry with all its departments.
- Created UIPath unattended robots to do many tasks such as logging in, multiple form-filling from an Excel spreadsheet, extracting information from websites and using it for different purposes. ([demo link](#))
- With the use of Studio, Orchestrator, and the robots, I was able to save the bank time and resources.
- Also spent a week in the Integration department and a week in the IT Governance and Risk Management.

GUC Innovators

Vehicle Dynamics Engineer

06/2024 – 07/2025

GUC Google Developer Groups

Robotics Engineer

10/2024 – 06/2025

German University in Cairo

Computer Science Junior Teaching Assistant

09/2024 – 01/2025

- Worked as a junior teaching assistant for CSEN102 and CSIS102 (Python) engineering and business informatics students in the GUC.
- Supervised by Assoc. Prof. Dr. Milad Ghantous.
- Gained many soft-skills such as leadership, communication, teaching, patience, accountability, and time management.

Computer Science Junior Teaching Assistant

02/2026 – 06/2026

- Worked as a junior teaching assistant for CSEN401 (OOP with Java) engineering students in the GUC.
- Helped student by conducting weekly office hours to help them on a project worth most of the grade.

Alstom

Engineering Internship

06/2024 – 08/2024

- Went through a rotation on many different engineering departments in the railway industry.
- Verification and Validation (V&V); learned about the different train supervision and testing systems such as the Automatic Train Supervision (ATS) and Automatic Train Control (ATC). I have conducted and analyzed signaling test cases on the line Umraniye-Atasehir Goztepe (UAG) in Turkey based on the client's needs, ensuring accurate train simulations within Alstom's Human Machine Interface (HMI) and effectively reported any discrepancies.

- Integrated Control Centre (ICC); acquired knowledge on SCADA, U400 by Alstom, virtualization, speed management and safe distancing, along with train scheduling and dispatching systems.
- Communication Based Interlocking (CBI); got presented to train interlocking structures like IXL and trackside equipment, such as point machines and balises, configuration rules, and all the trackside safety regulations. I have designed multiple railway tracks with signals and trackside equipment using AutoCAD, ensuring adherence to engineering rules and meeting Safety Integrity Level (SIL) 4 standards.
- Requirements and Configuration Management (RCM); was taught the different tendering processes in detail and projects' life cycles.
- I also got lectured on the ISTQB Certified Tester Foundation Level (CTFL) v4.0 where I gained an understanding of testing principles, methodologies, and tools, including test design techniques, static testing, and the role of testing throughout various software development lifecycles.
- Took lectures on cybersecurity and networks.

PROJECTS *(Available on GitHub)*

- University HR Management System ([GitHub Repository](#))** 12/2025
- Engineered a comprehensive relational database using SQL Server DBMS (including EERD and relational schema) with 40+ stored procedures to drive a responsive ASP.NET & JS fully functional website for multi-rank data management and automated payroll logic.
- Reviews data pipeline and sentiment API ([GitHub Repository](#))** 03/2026
- Built end-to-end Python ingestion and cleaning (Pandas), materialized relational SQLite tables from DDL scripts, staged deduplicated entities, appended review facts, and generated rolling sentiment aggregates with SQL analytics window functions.
 - Shipped HTTP services with FastAPI, Pydantic schemas, SQLite-backed queries, API key header authentication, and Uvicorn hosting, including readiness-style health verification against the database.
- Restaurant sales feature engineering and reporting ([GitHub Repository](#))** 03/2026
- Produced analytical datasets with Python, NumPy, and Pandas merges, grouping, temporal resampling to month-end aggregates, intentional missingness simulations, statistical imputation, and structured CSV publishing for downstream plotting.
 - Delivered exploratory and presentation outputs with Matplotlib trend charts, Seaborn correlation heatmaps on engineered metrics, and Plotly Express scatter plots exported to standalone HTML artifacts.
- Customer churn benchmarking on two tabular datasets ([GitHub Repository](#))** 03/2026
- Modeled churn with scikit-learn pipelines pairing StandardScaler, OneHotEncoder preprocessing, stratified splits, GridSearchCV, logistic regression benchmarking, persisted estimators (joblib) for reuse, and standard classification summaries.
 - Benchmarked nonlinear baselines (decision trees, random forests) against tuned logistic regression with cross-validated scoring, calibrated evaluation summaries, labeled confusion-matrix visualization (Matplotlib and Seaborn), and serialized best-performing models (joblib).
- Production-Ready Email AI Assistant ([GitHub Repository](#))** 04/2026
- Built an automated email assistant using LangGraph Cloud, featuring a React-based Chat UI that supports Human-in-the-loop interrupts and persistent thread management.
- Multimodal AI Chef Agent ([GitHub Repository](#))** 04/2026
- Developed a vision-integrated assistant using LangChain that utilizes short-term state management to provide recipe logic based on visual inputs.
- Context Aware AI Wedding Planner Agent ([GitHub Repository](#))** 04/2026
- Developed an automated coordinator using LangChain and Model Context Protocol (MCP) to manage multi-step logistics and maintain complex state across external databases.
- Travel Planning Web App ([GitHub Repository](#))** 11/2025
- Created a Node.js and Express server with a MongoDB NoSQL DBMS to implement secure session-based authentication, custom search algorithms, and database logic for managing personalized user collections.
- Jackaroo Computer Game ([GitHub Repository](#))** 05/2025
- Developed a multiplayer card game using JavaFX and Java 23, applying the MVC architecture for clean, modular design.
 - Built a responsive GUI with animation support and real-time interaction using asynchronous event handling.
- Advanced Persistent RDBMS Engine ([GitHub Repository](#))** 05/2026
- Custom Tiered Architecture: Engineered a high-performance Java database engine using a three-tier hierarchy (DBMS, Table, Page) with disk-based serialization and O(1) pointer-based record access.
- Von Neumann Architecture Simulator ([GitHub Repository](#))** 05/2026

- Created a cycle-accurate processor simulator in C, implementing a Von Neumann architecture with a custom 16-bit instruction set, unified memory space, and a Harvard-style hardware controller to manage instruction decoding and register-file operations.

GUC Project Portfolio Platform ([GitHub Repository](#)) 05/2026

- Architected a React and ASP.NET platform for GUC students to centralize lab work and bachelor projects, replacing scattered GitHub/Drive links with a unified academic portfolio.
- Engineered role-based workflows for students, instructors, and recruiters, administrators, featuring automated project discovery, internship matching, and structured feedback modules.

OS Simulator with GUI ([GitHub Repository](#)) 05/2026

- A simulated OS in C with a GTK3 GUI that manages processes in a 40-word RAM using PCB-based memory, disk swapping, and mutex synchronization, scheduled via Round Robin, HRRN, and MLFQ algorithms.

University Scheduling System ([GitHub Repository](#)) 04/2025

- Prolog-based class scheduling assistant that generates conflict-free student timetables, enforces two days off per week, and identifies available course slots.

Railway Network Pathfinder ([GitHub Repository](#)) 05/2025

- Haskell-based system using adjacency matrices and priority queues. Implemented Dijkstra's algorithm for shortest paths with dynamic edge updates and random network generation.

Audio Processing & Noise Cancellation System ([GitHub Repository](#)) 05/2025

- A Python solution using NumPy, Matplotlib, SoundDevice, and SciPy's FFT to generate multi-octave piano signals, synthesize a song via superposition, simulate noise with random sinusoids, and filter noise via frequency-domain analysis, restoring the original audio.

Band-Pass Filter Circuit ([GitHub Repository](#)) 05/2025

- Built a Sallen-Key filter in PSpice and breadboard using op-amps. Tuned resistors/capacitors for 1–100 kHz response, validated via AC/time-domain analysis and oscilloscope.

Basic Computer Design ([GitHub Repository](#)) 05/2025

- Logisim-built system with ALU, memory, registers, and control unit. Executed programs via opcode decoding and timing signals.

Professional Training, Certificates and Competitions

- Data Science, AI and ML Diploma (180 Hours) by Space Code Academy
- Data Science and AI Diploma (250 Hours) by Sprints.ai (In Progress)
- Foundation: Introduction to LangChain – Python by LangChain Academy
- Machine Learning Specialization by Stanford and DeepLearning.AI (In Progress)
- Statistical Data Analysis Using R Language by The City of Scientific Research and Technological Applications (SRTA-City)
- AI Data Analytics Intensive Bootcamp by EYouth
- McKinsey.Org Forward Program (In Progress)
- Introduction to Modern AI by Cisco
- L'Oréal Brandstorm 2025
- MATLAB Onramp by MathWorks
- Intro to AI by Microsoft and the Egyptian Ministry of Youth

Volunteering and Community

- Scouts Member: Ex-leader in cub and boy scouts | Current explorer scouts member. (2013 - Current)
- Organized and helped in multiple activities made for unprivileged Sudanese-Egyptian refugee kids such as charity graduation ceremonies and other fun days. (2023-2024)
- Organized and worked as one of the leaders in 2 kid summer camps made in Whippany, New Jersey. (06/2023 - 08/2023)

Other

- Egyptian Military Status: Exempted
- Dual Nationality: American-Egyptian